

Ring-Current Numerical Recipe

Supplemental method

We implemented a finite-polygon calculation of the field at target $\mathbf{r}$ from one aromatic source ring. For each conformation, `nmr_extract` performed the following steps.

1. Fit the ring geometry. Read the vertices

$\mathbf{v}_0, \dots, \mathbf{v}_{N-1}$ in canonical cyclic order and set their mean as the centre. Apply singular-value decomposition to the centred $N \times 3$ coordinate matrix. Its right singular vector for the smallest singular value is the plane normal, $\mathbf{n}$. If $\mathbf{n} \cdot [(\mathbf{v}_1 - \mathbf{v}_0) \times (\mathbf{v}_2 - \mathbf{v}_0)] < 0$, reverse $\mathbf{n}$.

2. Build the two wire paths. Coordinates

and the configured ring-specific displacement q are in Å. Set $\mathbf{p}_i^\pm = \mathbf{v}_i \pm q\mathbf{n}$. Join $\mathbf{p}_i^\pm$ to $\mathbf{p}_{(i+1) \bmod N}^\pm$, preserving the canonical direction and closing each path. Give each path 0.5 nA, for a 1-nA reference current. This discretises the two-path Johnson–Bovey construction on the observed polygon (Perkins et al. 1977, Case 1995).

3. Select a target. Query source rings within the configured calculation range. Reject a target–ring pair if the target is within the wire-distance guard of either path, belongs to the source ring, or is directly bonded to it.

4. Evaluate each directed edge. For each edge from $\mathbf{a}$ to $\mathbf{b}$, convert coordinates by $\mathbf{x}_m = 10^{-10}\mathbf{x}_\text{\AA}$ and current by $I_A = 10^{-9}I_{\text{nA}}$. Then set $\mathbf{d} = \mathbf{b} - \mathbf{a}$, $\mathbf{u} = \mathbf{r} - \mathbf{a}$, $\mathbf{v} = \mathbf{r} - \mathbf{b}$, and $\mathbf{c} = \mathbf{d} \times \mathbf{u}$. Evaluate the analytic straight-segment form (McGreivy et al. 2021) of the Biot–Savart line law (Haigh and Mallion 1979):

$$\mathbf{B}_{ab}(\mathbf{r}; I) = \frac{\mu_0 I}{4\pi} \frac{\mathbf{c}}{\|\mathbf{c}\|^2} \left(\frac{\mathbf{d} \cdot \mathbf{u}}{\|\mathbf{u}\|} - \frac{\mathbf{d} \cdot \mathbf{v}}{\|\mathbf{v}\|} \right).$$

Here I is the path current and μ_0 is the vacuum permeability. Return zero when $\|\mathbf{u}\| < \epsilon_{\text{end}}$, $\|\mathbf{v}\| < \epsilon_{\text{end}}$, or $\|\mathbf{c}\|^2 < \epsilon_{\text{axis}}$, using the configured guards.

5. Sum the field. Add $\mathbf{B}_{ab}$ over every edge of both paths, including their closing edges. Denote the result $\mathbf{B}^{(1\text{ nA})}$, and define the field per current as $\mathbf{b}_s = \mathbf{B}^{(1\text{ nA})}/(1\text{ nA})$. This analytic sum is the numerical field calculation; it uses no quadrature.

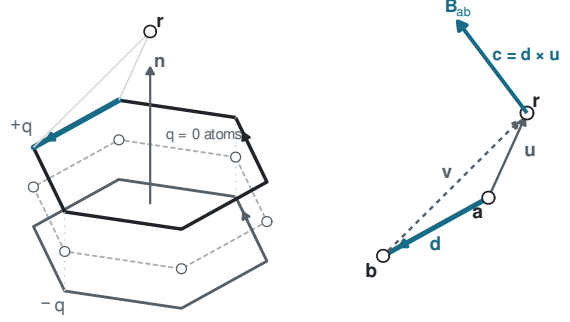


Figure 1: Finite-polygon construction. The observed ring is copied into two closed paths. The right panel shows one edge and its target vectors.

6. Package the field. For ring s , the induced current is $I_s = \kappa_s \mathbf{n}^\top \mathbf{B}_0$. The dot product is the normal component of the applied field. With the reference-current field already calculated, `nmr_extract` stores $\mathbf{G}_s = -10^6 \mathbf{b}_s \mathbf{n}^\top$ (Boyd and Skrynnikov 2002). The outer product packages the field; it does not generate it. The minus sign follows the shielding convention, and 10^6 is the parts-per-million factor. Decompose $\mathbf{G}_s$ into scalar T_0 , antisymmetric T_1 , and symmetric-traceless T_2 components and accumulate them by ring type. Multiplication by κ_s supplies the physical response downstream (Case 1995).

References

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